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A data reduction algorithm incorporating a low power continuous wavelet transform for use in wearable electroencephalography systems

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Chapter 1

Introduction

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Chapter 2

Literature review

2.1 Introduction

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2.2 Example display items

This is an example of providing a cross-reference to Chapter 3. Similarly, this is an example cross-reference to a sub-section, Section 3.2.

This is an example of adding references [1], [2], [3]. If you want the author name, or similar, you can use: A. J. Casson, E. Luna, and E. Rodriguez-Villegas in 2009 introduced a really good idea. (This is for when you primarily use numbered citations, but occasionally need an author's name. If using author names as the reference everywhere, change style=ieee in the biblatex setup above to whatever reference style you want, and then just use the cite command.)

For adding *emphasis* use the emph command or underscores such as *emphasis*.

An example table is given in Table 2.1. Note that the headings are inside a table.header environment to tell screen readers which cells are headers and which cells have the table content. Also, at the moment the template only adds the top horizontal line to the table. The others are added by hand in the table definition below. Ideally the template should detect

the end of the header, and the end of the table, and add these horizontal lines automatically, but this doesn't work yet.

Table 2.1. Probe results for design A.

Participant	Number (%)		Duration (%)	
	Prime dresses	Non-prime dresses	Prime dresses	Non-prime dresses
1	33.33	33.91	20.83	18.42
2	13.04	17.50	04.93	07.62
3	22.73	20.10	13.00	08.20
4	31.34	21.88	10.57	11.09
5	08.47	19.32	03.04	09.73
Mean	16.4	16.5	07.8	07.5
Standard deviation	09.7	06.6	05.4	03.3

This is an example equation in text $2 \sin \omega t$. (2.1) is an example of a displayed equation.

$$a^2 + b^2 = c^2 \tag{2.1}$$

Note that numbers are displayed differently in the text depending on how they are entered. Compare for example 123456 vs. 123456. Entering numbers directly, such as 1955, should be used for *text mode* numbers. That is, those representing text (dates, page numbers, and similar). Numbers representing maths, or variables or similar, should be entered inside $ $ so they are typeset in the same way as they appear in an equation. (This requires a bit of discipline, but helps ensure consistent use of number styles throughout.)

This is an example of a quote in text "*The electroencephalogram (EEG) is a classic non-invasive method for measuring a person's brainwaves*" [3]. Below is an example of a displayed quote.

> "*Electrodes are placed on the scalp to detect the microvolt-sized signals that result from synchronized neuronal activity within the brain.*" [3]

Fig. 2.1 is an example figure. Fig. 2.2 is an example of a figure made of sub-figures, using the subpar package, and a reference to one part of it looks like Fig. 2.2b.



Fig. 2.1. Example figure. Full caption goes here. A short caption is given as well, which goes in the list of figures to keep it tidy; it gets messy if there are long captions going over more than one line.



(a) (b) (c)

Fig. 2.2. Three copies of the University logo. (a) Copy one. (b) Copy two. (c) Copy three.

An example code listing is given below. Code in the body of the text can be included as `for` or `while` or `main`. This is just using the built in Typst functionality which is fairly limited. Could look at https://typst.app/universe/package/codly/ or similar to give more functionality such as line numbers, ability to link to a piece of code, and similar.

```
import numpy as np

def my_filter(in,f_obj):
    y = filter(f_obj,in)

    return y
```

2.3 Summary

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Chapter 3

Really good work

3.1 Introduction

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3.2 Content

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3.2.1 Introduction

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3.2.2 Detail

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3.2.3 More detail

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3.2.4 Summary

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3.3 Summary

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Chapter 4

Conclusions

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[2] A. J. Casson and E. Rodriguez-Villegas, 'Toward online data reduction for portable electroencephalography systems in epilepsy', *IEEE T. Biomed. Eng.*, vol. 56, no. 12, pp. 2816–2825, 2009.

[3] A. J. Casson, D. C. Yates, S. J. Smith, J. S. Duncan, and E. Rodriguez-Villegas, 'Wearable electroencephalography', *IEEE Eng. Biol. Med. Mag.*, vol. 29, no. 3, pp. 44–56, 2010.

Appendices

Appendix A

First Appendix

A.1 Section in Appendix



Fig. A.1. Example figure in Appendix.

Appendix B

Second Appendix

B.1 Section in Appendix